

Diabetic Retinopathy Grading from Fundus Photographs: An Ordinal Classification Approach using Deep Learning

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Abstract

1 Introduction

Diabetic Retinopathy (DR) is a complication of diabetes and a leading cause of preventable blindness. It is graded on a five-point scale, from no retinopathy (Grade 0), through mild, moderate, and severe non-proliferative DR, to proliferative DR (Grade 4). Early stages are asymptomatic, so screening relies on colour fundus photography, where clinicians look for microaneurysms, haemorrhages, exudates, and neovascularisation. Manual grading is slow, requires specialist expertise, and varies between graders, which has motivated automated deep-learning systems that grade fundus images directly.

For this work we use the publicly available **APTOS 2019 Blindness Detection** dataset (Kaggle), containing 3,662 high-resolution colour fundus photographs from rural India, labelled by clinicians on the five-class International Clinical Diabetic Retinopathy (ICDR) severity scale (0–4).

2 Problem Statement

The task of DR grading from fundus photographs is fundamentally an *ordinal* five-class classification problem and the cost of misclassification grows with the distance between the predicted and true grade. Standard image classification models that treat the grades as independent categorical labels therefore discard valuable information and tend to produce clinically implausible errors. The problem is further complicated by a severe class imbalance, since the majority of patients exhibit no or mild disease, leaving the higher-severity grades drastically under-represented in available datasets. The objective of this work is to design a convolutional neural network that perform ordinal five-class grading upon classification, mitigates class imbalance, and produces interpretable, clinically aligned predictions evaluated through the Quadratic Weighted Kappa (QWK) metric and grad-CAM analysis to identify which regions of the image are most important for each prediction.

2.1 Baseline Model

We approached the problem the way any standard image classification task would be approached: with an ImageNet-pretrained CNN, a linear head re-mapped to the five DR grades, and a vanilla cross-entropy loss. Three backbones of progressively increasing capacity were chosen so that we could observe how depth and architectural sophistication interact with our small ($\sim 3.6k$ -image) and heavily imbalanced training set. The first model, used as our reference baseline, is **ResNet-18** ($\sim 11.2M$ parameters)—a residual network deep enough to learn rich

features but light enough to train end-to-end on a single 4 GB consumer GPU. The second is **ResNet-50** (~ 25 M parameters), chosen to test whether simply going deeper improves performance under identical training conditions. The third is **EfficientNet-B0** (~ 5 M parameters), selected as a more parameter-efficient, compound-scaled architecture that often dominates the medical imaging literature for fundus tasks. Across all three runs the optimisation recipe was kept deliberately fixed—Adam optimiser at $\text{lr} = 10^{-4}$, 224×224 input resolution, 10 training epochs, an 80/20 stratified train-validation split, and unweighted **categorical cross-entropy** as the loss function—so that any differences in performance can be traced back to the architecture rather than the optimisation strategy.

2.2 Baseline Results

The ResNet-18 baseline achieved a best validation Quadratic Weighted Kappa of **QWK** = 0.8819 and accuracy of **83.49%**, meeting standard performance benchmarks. However, the core finding from the baseline experiments reveals a critical failure: **severe overfitting and class imbalance**. The training and validation curves (Figure 1) clearly show the divergence between training and validation loss over 10 epochs, indicating the model is memorising patterns rather than learning generalizable features.

Table 1: ResNet-18 Baseline Classification Report

Grade	Precision	Recall	F1-Score	Support
Grade 0	0.94	0.96	0.95	361
Grade 1	0.50	0.04	0.07	74
Grade 2	0.57	0.94	0.71	200
Grade 3	0.64	0.18	0.28	39
Grade 4	0.72	0.22	0.34	59
Accuracy	0.76			733

The per-class classification performance (Table 1) exposes the clinical inadequacy of this approach. While Grade 0 (healthy) is recognised almost perfectly (F1-score = 0.95), the minority and clinically critical grades severely degrade: Grade 3 and Grade 4 show F1-scores of only 0.28 and 0.34 respectively. The model fails where it matters most—on advanced disease stages. This pattern confirms that standard categorical cross-entropy loss and naive training are fundamentally unsuited to this ordinal, imbalanced medical imaging task.

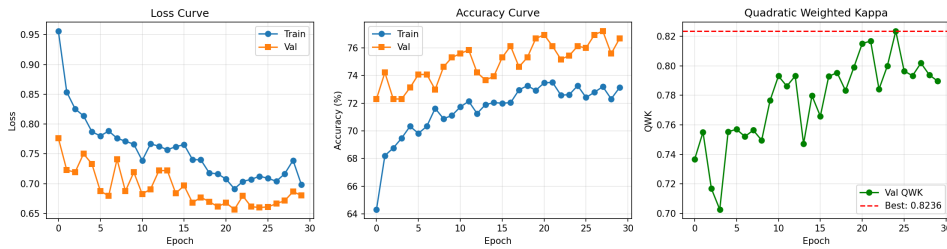


Figure 1: Training and validation loss curves over 10 epochs, showing characteristic overfitting behaviour.

2.3 Defining Objectives

Through baseline experiments, it is evident that *any* typical image classification CNN, when transplanted directly onto APTOS-2019, fails in the same predictable way: it overfits a small, heavily class-imbalanced medical dataset differ sharply from the natural-image distribution the

backbone was pretrained on. The objectives of this project therefore reorient away from chasing yet another architecture and towards directly attacking the root causes:

1. **Reduce overfitting** through processes like data augmentation, regularisation (dropout, weight decay), and stronger normalisation of the train/validation curves.
2. **Fine-tune ResNet and find an appropriate loss function** that better capture the ordinal nature of the problem, such as focal loss and CORAL loss that explicitly models the rank order of the classes.
3. **Explore alternatives to flat softmax classification** that respect the ordinal structure and class imbalance of DR severity, including weighted cross-entropy,

3 Literature Review

3.1 Model Selection

DR classification has largely tracked the broader ImageNet backbone progression. Gulshan *et al.* [1] set the template with an **Inception-v3**-based binary referable-vs-non-referable classifier that reached ophthalmologist-level sensitivity on 128k images, after which work on five-class severity grading shifted to deeper residual networks (**ResNet-50**, **DenseNet-121**) and compound-scaled **EfficientNet** variants. Across these systems the classification head is uniformly a flat softmax over Grades 0–4, which discards the ordinal structure and penalises a Grade-4-as-Grade-0 confusion no more than a Grade-4-as-Grade-3 one—an assumption consistent with the high inter-grader disagreement reported on the same images by human ophthalmologists [2].

For our setting—a ~ 3.6 k-image training set on a single 4 GB consumer GPU, with ordinality to be supplied by the loss rather than the backbone—the relevant axis is not raw capacity but the smallest backbone that still gives useful ImageNet-pretrained features without overfitting. **ResNet-50** (~ 25 M parameters) brings roughly $2.2\times$ the parameters of ResNet-18 and, at this data scale, simply enlarges the overfitting surface; **EfficientNet-B0** (~ 5 M parameters) is parameter-efficient on paper but its compound-scaling advantage requires more data than we have to be realised. **ResNet-18** (~ 11.2 M parameters) sits at the sweet spot—deep enough to learn the multi-scale lesion textures that distinguish DR severity grades, shallow enough to fine-tune end-to-end in memory, and well-matched to the data regime so that gains from regularisation and ordinal-aware losses are not masked by capacity-driven overfitting. We therefore adopt **ResNet-18** as the backbone and direct the remaining effort towards the classification head, the loss function, and the preprocessing pipeline.

3.2 Preprocessing

Preprocessing is not a cosmetic step in medical imaging, it is the single largest lever between a model that learns lesion-level pathology and one that latches onto background artefacts. Raw fundus photographs vary enormously in illumination, contrast, white-balance, the size of the visible retinal disc, and the amount of black border surrounding it; without standardisation, the network is forced to spend representational capacity on these nuisance factors instead of on disease cues [13, 14]. A consistent finding in the medical CNN literature is that, in the absence of preprocessing, gradient-based attribution maps drift onto clinically irrelevant regions (background, large vessels, image borders), whereas appropriate preprocessing concentrates attention on the diagnostic structures themselves [13].

Within our pipeline, preprocessing is built around four mutually reinforcing components. **Geometric normalisation** resizes every image to a fixed 224×224 resolution and crops the circular field of view, removing the black corners that otherwise contribute uninformative gradients.

Photometric normalisation subtracts the per-channel ImageNet mean (0.485, 0.456, 0.406) and divides by its standard deviation (0.229, 0.224, 0.225), ensuring that the feature statistics fed to the pretrained ResNet match those it was trained on, without this step, transfer learning is markedly less effective. **Data augmentation** (horizontal/vertical flips, small rotations, brightness-contrast jitter, CLAHE contrast enhancement) synthetically expands the effective training distribution and is one of the most powerful regularisers known for limited medical datasets [12, 7]. **Regularisation hyperparameters** that complement preprocessing, most notably *dropout*, which randomly disables units during training to prevent co-adaptation [10], and *weight decay*, which penalises large parameter norms, directly address the overfitting we observed in the baseline. Closely related are the choice of *weight initialisation* [11], the optimiser learning rate, and the use of pretrained ImageNet weights.

3.3 Loss Function Definition

Initially, the most obvious choice for a loss function is the standard **categorical cross-entropy**, which treats the five DR grades as independent classes. However, this approach discards the ordinal structure of the problem and fails to account for class imbalance, later introduced was **weighted cross-entropy** in [6], which assigns higher loss weights to the minority classes (Grades 3 and 4) to mitigate imbalance. **Focal loss** [5] further refines this by down-weighting easy examples and focusing the model’s attention on hard, misclassified samples, which is particularly beneficial in imbalanced settings. This brings our class imbalance down to a manageable level, but it still treats the classes as independent. **Ordinal regression losses** such as CORAL [9] explicitly model the rank order of the classes by decomposing the problem into a series of binary classification tasks, ensuring that the model learns to respect the natural ordering of DR severity. This handles the severity of misclassification, for example, misclassifying Grade 4 as Grade 0 is penalised more than misclassifying it as Grade 3. This is a critical improvement over standard cross-entropy, which treats all misclassifications as equally bad.

3.4 Evaluation Metric

Plain accuracy is poorly suited to a five-class, severely imbalanced, ordinal task: predicting Grade 0 for every image already attains close to 50% accuracy on APTOS-2019 without learning anything clinically useful. We therefore adopt the **Quadratic Weighted Kappa (QWK)** as the primary metric, following Cohen’s weighted-kappa framework [15], which rewards agreement between predicted and true grades and penalises disagreement by the *squared* distance between them, so a Grade-4-as-Grade-0 error is sixteen times costlier than an adjacent-grade error. QWK is the official scoring metric of both the Kaggle *Diabetic Retinopathy Detection* (2015) and *APTOS-2019 Blindness Detection* competitions, and de la Torre *et al.* [16] have shown that it doubles as a differentiable training loss for DR grading.

We supplement QWK with overall **accuracy** (for comparability with prior reports, while remembering its insensitivity to imbalance [6]), **macro-F1** (equally-weighted per-class F1, which surfaces collapses on minority Grades 3 and 4), and the full **confusion matrix** (which reveals *which* grades are being conflated and is what exposed the baseline’s collapse onto Grades 0 and 2).

4 Methodology

5 Results

6 Future Directions

7 Conclusion

References

- [1] V. Gulshan *et al.*, “Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs,” *JAMA*, vol. 316, no. 22, pp. 2402–2410, 2016.
- [2] J. Krause *et al.*, “Grader Variability and the Importance of Reference Standards for Evaluating Machine Learning Models for Diabetic Retinopathy,” *Ophthalmology*, vol. 125, no. 8, pp. 1264–1272, 2018.
- [3] H. Fu, M. Gong, C. Wang, K. Batmanghelich, and D. Tao, “Deep Ordinal Regression Network for Monocular Depth Estimation,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 2002–2011.
- [4] Z. Niu, M. Zhou, L. Wang, X. Gao, and G. Hua, “Ordinal Regression with Multiple Output CNN for Age Estimation,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 4920–4928.
- [5] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal Loss for Dense Object Detection,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2980–2988. [Online]. Available: <https://arxiv.org/pdf/1708.02002v2>
- [6] M. Buda, A. Maki, and M. A. Mazurowski, “A Systematic Study of the Class Imbalance Problem in Convolutional Neural Networks,” *Neural Networks*, vol. 106, pp. 249–259, 2018.
- [7] L. Perez and J. Wang, “The Effectiveness of Data Augmentation in Image Classification using Deep Learning,” *arXiv preprint arXiv:1712.04621*, 2017.
- [8] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 618–626.
- [9] W. Cao, V. Mirjalili, and S. Raschka, “Rank Consistent Ordinal Regression for Neural Networks with Application to Age Estimation,” *Pattern Recognition Letters*, vol. 140, pp. 325–331, 2020. [Online]. Available: <https://sebastianraschka.com/pdf/papers/2019-coral1-2019-arxiv.pdf>
- [10] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, “Dropout: A Simple Way to Prevent Neural Networks from Overfitting,” *Journal of Machine Learning Research*, vol. 15, no. 56, pp. 1929–1958, 2014. [Online]. Available: <https://jmlr.org/papers/v15/srivastava14a.html>
- [11] K. He, X. Zhang, S. Ren, and J. Sun, “Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2015, pp. 1026–1034.

- [12] C. Shorten and T. M. Khoshgoftaar, “A Survey on Image Data Augmentation for Deep Learning,” *Journal of Big Data*, vol. 6, no. 60, pp. 1–48, 2019. [Online]. Available: <https://link.springer.com/article/10.1186/s40537-019-0197-0>
- [13] A. W. Salehi *et al.*, “Medical Image Analysis Using Deep Learning Algorithms,” *Frontiers in Public Health*, vol. 11, 2023. [Online]. Available: <https://www.frontiersin.org/journals/public-health/articles/10.3389/fpubh.2023.1273253/1>
- [14] G. Litjens *et al.*, “A Survey on Deep Learning in Medical Image Analysis,” *Medical Image Analysis*, vol. 42, pp. 60–88, 2017.
- [15] J. Cohen, “Weighted Kappa: Nominal Scale Agreement Provision for Scaled Disagreement or Partial Credit,” *Psychological Bulletin*, vol. 70, no. 4, pp. 213–220, 1968.
- [16] J. de la Torre, D. Puig, and A. Valls, “Weighted Kappa Loss Function for Multi-class Classification of Ordinal Data in Deep Learning,” *Pattern Recognition Letters*, vol. 105, pp. 144–154, 2018.